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SUPERSEDING
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MILITARY SPECIFICATION

WELDING MACHINES, RESISTANCE, SPOT, PORTABLE TYPE, SINGLE PHASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, resistance, spot welding machines of the single phase direct energy type with equipment and accessories specified herein.

1.2 Classification. The welding machines shall be of the following sizes. The size to be furnished shall be as specified (see 6.2.1).

Size 1 - 5 KVA transformer rating.
Size 2 - 10 KVA transformer rating.
Size 3 - 15 KVA transformer rating.
Size 4 - 20 KVA transformer rating.
Size 5 - 30 KVA transformer rating.
Size 6 - 40 KVA transformer rating.
Size 7 - 50 KVA transformer rating.
Size 8 - 75 KVA transformer rating.
Size 9 - 100 KVA transformer rating.
Size 10 - 125 KVA transformer rating.
Size 11 - 150 KVA transformer rating.
Size 12 - 200 KVA tramsformer rating.

1.2.1 Machine size. The welding machine size is expressed in terms of kilovolt ampere (KVA) at 50 percent duty cycle of the welding transformer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-W-12332 - Welding, Resistance, Spot, Seam, and Projection, for Fabricating Assemblies of Low-carbon Steel.

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, CHAPTER XVII, PART 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951- Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 5 - Resistance Welding Control.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA - 56 - Resistance Welding Manual Volume 1.

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards.

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103-1403.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding machine shall be new and one of the manufacturer's current models capable of producing commercial quality spot welds on ferrous metal in accordance with the requirements herein. The welding machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

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3.2.1 Dimensional systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the welding machine. When only one system dimensional is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These measurements may by be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the welding machine shall be graduated in the same system. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the welding machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the welding machine.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the welding machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the welding machine, whether or not specifically required herein, shall comply with all of the requirements of CFR 29, Chapter XVII, Part 1910 that are applicable to the machine itself. The welding machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial welding machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail (see 6.2.1).

3.2.5 Mercury restriction. The welding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

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3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding machine.

3.2.7 Environmental protection. The welding machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.8 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.9 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or welding machine.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the welding machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28.

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3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose.

3.4 Components. Unless otherwise specified (see 6.2.1), the welding machine shall consist of two basic units connected by power cables and when required a pressure supply hose. One unit shall consist of an enclosure, electrical system, a cooling system, and a pressure supply system. The other unit shall consist of the welding gun assembly.

3.4.1 Enclosure. The enclosure shall be constructed of corrosion resistant metal or metal treated to resist corrosion. Ventilation shall be adequate to prevent temperatures in excess of design requirements under normal operating conditions of rated power output. Enclosed components shall be accessible for inspection and maintenance. Doors and access panels shall be provided with locks and interlocks to prevent access by unauthorized personnel to live portions of the machine.

3.4.2 Electrical system. The electrical system shall include the welding transformer, rectifiers, timers, meters, circuit protection, disconnect switches, and all controls required for operation. The electrical system and components shall conform to the applicable requirements of NFPA 70. Each sub-panel shall be arranged so that it may be removed independently of any other sub-panel. All terminals and units shall be clearly indentifiable from the electrical wiring diagram furnished with the machine. Where ventilation is provided by forced air, the air inlet shall be provided with replaceable filters.

3.4.2.1 Electrical service requirements. The welding machine shall operate on a single phase, 60 hertz (Hz), primary electric service at either 220 or 440 volts. Unless otherwise specified (see 6.2.1), the machine shall be provided to operate on 440 volts. All flexible power supply cables shall be reinforced or armored at machine cable junction.

3.4.2.2 Main disconnect switch. The welding machine shall have a main fusible disconnect switch or circuit breaker. Unless otherwise specified (see 6.2.1),a circuit breaker shall be mounted on the machine panel.

3.4.2.3 Solid state components. Solid state design shall be used throughout for electronic components. Use of relays shall be only as required to assure adequate safety and reliable operation.

3.4.2.4 Welding transformer. The welding transformer shall be of the overhead balanced beam mounted type. The transformer shall be securely clamped and mounted to eliminate shifting due to shock, stresses, vibration, or other conditions encountered during operation. The transformer KVA rating and secondary current output shall be in accordance with table I for the machine size specified. When the transformer is water cooled it shall be thermostatically controlled to minimize condensation. The transformer insulation shall conform to RWMA Bulletin Number 16 class B or better insulation. A temperature responsive device shall be provided to prevent energization of the transformer, if the temperature exceeds design limits. When specified (see

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6.2.1), the transformer shall have a series/parallel switch in the primary circuit. A grounding reactor shall be provided on the secondary circuit.

TABLE I. Machine size and capacity.

Machine size	Welding transformer power rating (KVA)	Electrode force (pounds)	Secondary short circuit current 1/ (amperes)	Weld capacity combined thickness of SAE 1010 steel (inches)	Low reactance cable size 6 ft long 2/
1	5	25	3,000	.008	2 AWG
2	10	25	4,000	.015	2 AWG
3	15	25	6,000	.050	0 AWG
4	20	300	9,000	.042	257 MCM
5	30	400	10,000	.062	267 MCM
6	40	500	11,000	.068	267 MCM
7	50	600	12,000	.080	300 MCM
8	75	800	14,000	.120	300 MCM
9	100	1000	16,000	.130	300 MCM
10	125	1500	17,000	.156	450 MCM
11	150	2000	19,000	.188	450 MCM
12	200	3000	22,000	.218	450 MCM

NOTE: Requirements are not less than those shown.
The pneumatic or pneumatic-hydraulic system being supplied with air pressure of 80 psi.
1/ Amperes + 15 percent.
2/ AWG - American Wire Gage. MCM - Thousand Circular Mil.

3.4.2.5 Electrical controls. The electrical controls shall consist of the welding power control, timing and sequence control, current adjustment control, welding contactor, and auxiliary and accessory controls. Operator controls shall be accessible to the operator. All calibrations, phase shift controls, and similar characteristics shall be clearly marked in standard terms.

3.4.2.5.1 Operator control panel. The operator control panel shall have an on-off or weld-no weld switch, a power "on" and sequence lights, a current adjustment control, a weld timer selector, auxiliary and accessory control function switches, and any other controls or switches required to meet the requirements specified herein. When furnished, all meters and gauges shall have an accuracy of ± 2 percent at full scale reading.

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3.4.2.5.2 Welding power control. Unless otherwise specified (see 6.2.1), the welding machine shall be furnished with a synchronous precision combination welding power control unit conforming to NEMA ICS 5, type S2H mounted as an integral part of the machine.

3.4.2.5.3 Timing and sequence controls. The control equipment shall provide operating and adjustment of the welding variables to ensure that optimum values of force and current timing and sequencing can be repeatitively attained to produce uniform spot welds in accordance with this specification. Unless otherwise specified (see 6.2.1), the welding machine shall be furnished with a sequence timer conforming to NEMA ICS 5, type 7B. The timing control shall be calibrated in cycles based on the electric service line frequency of 60 Hz. The timer shall have not less than four functions: squeeze, weld, hold, and off. When specified (see 6.2.1), additional auxiliary control functions shall be furnished. The controls shall have signal lights on the panel to indicate the operating sequence of the welding cycle and to aid in trouble shooting when a malfunction occurs. Relays within control circuits shall be of the plug-in type. Timing and sequencing functions shall be grouped on a plug-in chassis. Provisions shall be made for the addition or removal of auxiliary functions. The control chassis and components shall be mounted in such a manner as to assure long service life and ease of maintenance and replacement.

3.4.2.5.4 Current adjustment control. The current adjustment control shall provide continuously variable adjustment of the welding current from the upper to the lower limits of the machine capacity. The control shall be graduated in increments of 1 from 0 to not less than 99 or from 10 percent to not less than 99 percent for convenience in identifying settings. The current adjustment control shall be such that, at the maximum setting of the adjustment, the maximum short-circuit current shall be in accordance with table I for the machine size specified.

3.4.2.5.5 Welding contactor. The welding contactor shall be a silicon controlled rectifier (SCR) type. The SCR contactor shall be thermostatically protected and shall be mounted in a heat sink. Machine size 5 through size 12 shall be water cooled and protection shall be provided to prevent welding if the water supply is shut off or restricted. The SCR contactor shall be protected from excessive power demands.

3.4.2.5.6 Automatic voltage compensator. When specified (see 6.2.1), an automatic voltage compensator (AVC) shall be provided to minimize the effects of line voltage variations. The AVC shall maintain the welding current within ± 2 percent for a line voltage variation of ± 10 percent. Based on the input line frequency, full compensation shall occur within 1-1/2 cycle for a sudden change in line voltage.

3.4.2.5.7 Control circuit voltage. Control equipment shall be capable of operating on control circuit voltage which varies as much as ± 10 percent from the adjusted voltage. All switch inputs shall be isolated from the input electric power supply by a low voltage transformer having a secondary voltage not to exceed 24 volts.

3.4.2.5.8 Accessory controls. Accessory controls shall consist of not less than the controls specified in 3.4.2.5.8.1 through 3.4.2.5.8.3.

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3.4.2.5.8.1 Weld on-off or weld-no weld switch. A weld-no weld switch shall be provided. In the "off" or "no weld" position, the control sequence will actuate with no welding current flow.

3.4.2.5.8.2 Repeat on-off or repeat-non-repeat switch. A repeat on-off switch shall be provided. In the "off" or "non-repeat" position only one welding cycle will result with the actuation of the welding switch. In the "on" or "repeat" position the welding cycle will repeat automatically after a controlled off time interval as long as the switch remains closed.

3.4.2.5.8.3 Actuation switch. The actuation switch shall control the electrode assembly and, when applicable, its associated pneumatic and hydraulic system and initiate the welding cycle.

3.4.2.6 Electrodes and electrode holders. The electrodes and electrode holders shall conform to RWMA Bulletin Number 16. The quantity, size, type, and alloy class of the electrodes shall be as specified (see 6.2.1).

3.4.3 Electrode force system. Machine sizes 1 and 2 shall be furnished with a manual electrode force system. Machine sizes 3 through 12 shall be furnished with a pneumatic or a pneumatic hydraulic electrode force system as specified (see 6.2.1). The electrode force of each system shall be adjustable and shall provide a means of locking at the selected force. Unless otherwise specified (see 6.2.1), the gun provided shall be a direct action, C-type, with a throat depth of 8 inches. The gun working and retracting stroke and the bail suspension system shall be as specified (see 6.2.1).

3.4.3.1 Manual electrode system. The manual system shall provide for single hand operation. The force actuation lever shall be located on the upper face of the gun.

3.4.3.2 Pneumatic system. The pneumatic system and components shall conform to the requirements of RWMA Bulletin Number 16, where applicable, and not in conflict with the requirements of this specification.

3.4.3.2.1 Pneumatic system operation. Unless otherwise specified (see 6.2.1), the pneumatic system air cylinder shall be a double action type and shall be designed to operate with 80 pounds per square inch (psi) air pressure input service. The inlet connection shall be marked for the pressure requirements. The pneumatic system shall include a filter, valving, piping, regulators, gauges, lubricators, moisture drain provisions, wiring, switches, and all related equipment necessary to operate the welding gun. The pneumatic system shall be attached to the transformer unit. The system shall be controlled by a solenoid valve actuated by a trigger or push-button located on the gun handle. The gun to cable connections shall be adaptable for a concentric, low reactance, water or air cooled type of cable.

3.4.3.2.2 Air gauges. The air gauges shall have a face diameter of not less than 4-1/2 inches. The gauges shall be calibrated in psi with an accuracy within + 0.5 percent of the maximum reading of the gauge. Means of shock protection shall be provided to prevent gauge damage. The gauge shall be mounted on the front or side of the machine and be visible to the operator.

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3.4.3.2.3 Air regulators. The air regulators shall be self-bleeding. The regulators and solenoid valves shall be installed with unions. The regulators shall maintain the output pressure at ± 0.5 psi as long as the supply lines remains at least 2 psi higher than the preset pressure regulator throughout the full range of 60 to 100 psi. The air system shall be capable of producing the electrode force in accordance with table I for the machine size specified.

3.4.3.3 Pneumatic-hydraulic force system. The pneumatic part of the pneumatic-hydraulic system shall be in accordance with the requirements of 3.4.3.2. The hydraulic booster system shall be equipped with a double acting piston, a self-bleeding system, an oil reservoir, a means to check fluid level, and a data plate citing fluid quantity and type to be utilized in the hydraulic system. Hydraulic lines, fittings, and components shall be capable of withstanding twice the maximum operating pressure without damage or failure. When specified (see 6.2.1), heaters or coolers shall be furnished for the hydraulic system to meet any unusual climatic operational requirements.

3.4.4 Cable and lines. The water, pneumatic, and hydraulic lines utilized to connect the welding gun to the machine shall be clamped at least every three linear feet or shall be molded integral with the weld current cable. Weld current cable shall be of the concentric low reactance, water or air cooled type as specified (see 6.2.1). Weld cable current rating shall exceed maximum weld current by not less than 20 percent. Unless otherwise specified (see 6.2.1), cables and lines shall be six feet in length. Cable and line external covering shall resist abrasion, impact, tearing, burning , oil absorption, and corrosion. Fluid lines shall be of the quick disconnect type and shall not require the use of hand tools for connection to machine or gun.

3.4.5 Cooling systems. The cooling system for machine sizes 1 through 4 shall be either air or water type as specified (see 6.2.1). Machines larger than size 4 shall be water cooled. Cooling systems shall be adequate to meet requirements contained herein without exceeding design temperature or causing a hazard to the operator.

3.4.5.1 Air cooling system. Air cooling system utilizing blower type forced air cooling shall be equipped with replaceable inlet air filters. A pilot light shall indicate cooling system activation.

3.4.5.2 Water cooling system. When supplied with water in accordance with the recommendations of the manufacturer, the water cooling system will operate to prevent the temperature rise during operation from exceeding the limits specified in RWMA Bulletin Number 16. The system shall operate from a shop tap water supply, and shall have a single inlet connection into a manifold. The inlet connection shall be marked for pressure and flow requirements. The system shall have parallel flows from the manifold to the welding transformer, to the secondary winding and circuit conductors, to the primary welding current contactors and to the upper and lower electrodes. The piping in the system shall be corrosion-resistant to tap water service. The system shall have an inlet strainer, and a closed manifold drain system with flow indicators. The water flows lines to the electrodes shall have solenoid valves to prevent water flow while changing the electrodes. When specified (see 6.2.1), the system shall have water saving devices.

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3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the welding machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. The welding machine shall be capable of producing spot welds under production conditions where portability is a requirement. The welds shall meet the requirements of MIL-W-12332.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial welding machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to each welding machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A corrosion-resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below and that information required in RWMA Bulletin Number 16 for nameplate data. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Water pressure required
Temperature of input cooling water
Date of manufacture
Contract Number or Order Number
National Stock Number or Plant Equipment Code

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3.11 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the welding machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance

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inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The welding machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The welding machine shall be operated at no load for not less than 15 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance welding test. The welding machine shall be tested for welding capability by performing thirty consecutive welds using two thicknesses of the material specified in table I. The welds shall be examined and tested in accordance with MIL-W-12332.

4.7.3 Automatic voltage compensator test. When the AVC is specified (see 3.4.2.5.6), the performance shall be measured by means of a calibrated recording instrument. With the welding current adjustment set first at 50 percent and them to 90 percent of maximum, the secondary current and primary voltage shall be recorded as ± 10 percent step changes are introduced into the primary voltage. Alternately, the compensation and response time may be tested by causing the voltage change to occur only on the control of the compensator, in which case, the 10 percent increase or decrease in voltage shall cause, respectively, a decrease or increase of 10 percent in the secondary current within ± 2 percent. The secondary current compensation and response time shall be as specified in 3.4.2.5.6.

4.7.4 Dielectric test. The welding machine shall be subjected to the dielectric test as described in RWMA Bulletin Number 16.

4.7.5 Electrode force test. The maximum electrode force shall be measured by means of an electrode force gauge. The value shall be not less than that specified in table I.

4.7.6 Rated capacity test. The welding machine shall be adjusted to weld two equal thicknesses of SAE 1010 low carbon steel for the machine size specified. The test shall be conducted without any material between the electrode tips and with the current control adjusted so that the secondary current under the short-circuit condition is equal to the current used for welding the material. The off-time shall be set for the maximum welds per minute in accordance with RWMA-56, Volume 1. The welding machine shall be operated for a period not less than 1 hour without readjustment of the controls or relays. During this test, the secondary current flow time and amplitude shall be measured by means of a calibrated recording instrument for 2 successive periods of flow at the beginning of the test and at the end of the test. The variation in the peak value of the welding current shall not exceed 5 percent. The current flow time shall not be variable. The equipment may be stopped for 1 period not to exceed 5 minutes, if necessary, to dress the electrode tips. At the end of the test the equipment shall be shut down

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and all external parts shall be examined for overheating. The temperature limits specified in RWMA Bulletin Number 16 shall not be exceeded. If the equipment fails to conform to any of these requirements, or fails to operate properly during the test, the test shall be discontinued and the supplier permitted to make any necessary adjustments. After adjustments, the test in which the equipment failed shall be rerun. If the equipment fails the rerun test, failure shall be cause for rejection.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The equipment covered by this specification shall be capable of producing spot welds in accordance with this specification under production conditions and is primarily intended for producing spot welds of a peel test quality in mild steel for applications where the portability of electrode spot welders facilitates weldment fabrication.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size welding machine required (see 1.2).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US or SI), if required (see 3.2.1.1).

g. Additional safety and health requirements, if required (see 3.2.4).

h. Painting, if different (see 3.3.5).

i. Components, if different (see 3.4.).

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j. Electrical system and input power, if different (see 3.4.2.1).

k. Circuit breaker and its mounting, if different (see 3.4.2.2).

l. Series/parallel switch, if required (see 3.4.2.4).

m. Welding power control, if different (see 3.4.2.5.2).

n. Timing and sequencing control, if different (see 3.4.2.5.3).

o. Auxiliary control functions, if required (see 3.4.2.5.3).

p. Automatic voltage compensator, if required (see 3.4.2.5.6).

q. Quantity, size, type, and alloy class of electrodes required (see 3.4.2.6).

r. Specify type of force system required (see 3.4.3).

s. Type gun and throat depth, if different (see 3.4.3).

t. Specify gun stroke and bail suspension system required (see 3.4.3).

u. Pneumatic system air cylinder, if different (see 3.4.3.2.1).

v. Hydraulic heaters or coolers, if required (see 3.4.3.3).

w. Specify type weld current cables required (see 3.4.4).

x. Cable and length, if different (see 3.4.4).

y. Specify type cooling system required (see 3.4.5).

z. Water saving devices, if required (see 3.4.5.2).

aa. Size and capacity, if different (see 3.5).

bb. Additional performance requirements, if required (see 3.6.1).

cc. Standard equipment, if different (see 3.7).

dd. Specify and fully describe optional equipment, if required (see 3.8).

ee. Specify applicable repair parts, if required (see 3.9).

ff. First article inspection, if different (see 4.4)

gg. Quality conformance inspection, if different (see 4.5).

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hh. If preservation and packing in accordance with MIL-W-45562 is required, specify level of preservation and packing (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Automatic voltage compensator
Resistance welder
Sequence timer
Spot welder
Synchronous welding power control
Welding gun, direct action (C-type)
Welding gun, pincher type
Welding machine, resistance
Welding machine, spot

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Air Force - 99

Preparing activity:
DLA - IP

(Project 3432-0136)

Review activities:
Navy - MC, AS, OS
Air Force - 84

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80121B	Welding Machines, Resistance, Spot, Portable Type, Single Pha

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 82 MAR PREVIOUS EDITION IS OBSOLETE.

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